

Data Preprocessing Report

Loan Eligibility Preprocessing & Feature Engineering

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A. Dataset Inspection Report

1. Loading the Dataset

The Loan Prediction dataset using `pd.read_csv()`. The file loaded cleanly, with all 13 columns parsed and no encoding or delimiter issues.

2. First 10 Rows (`df.head(10)`)

Loan_ID	Gender	Married	Dep.	Education	Income	LoanAmt	Status
LP001002	Male	No	0	Graduate	5,849	– (missing)	Y
LP001003	Male	Yes	1	Graduate	4,583	128	N
LP001005	Male	Yes	0	Graduate	3,000	66	Y
LP001006	Male	Yes	0	Not Graduate	2,583	120	Y
LP001008	Male	No	0	Graduate	6,000	141	Y
LP001011	Male	Yes	2	Graduate	5,417	267	Y
LP001013	Male	Yes	0	Not Graduate	2,333	95	Y
LP001014	Male	Yes	3+	Graduate	3,036	158	N
LP001018	Male	Yes	2	Graduate	4,006	168	Y
LP001020	Male	Yes	1	Graduate	12,841	349	N

(Table condensed for readability; *CoapplicantIncome*, *Loan_Amount_Term*, *Credit_History*, and *Property_Area* are also present in the full dataset.) Note row 1 already shows a missing *LoanAmount* value, an early signal that missing data will need to be addressed.

3. Dataset Structure

Property	Value
Rows (loan applications)	614
Columns (variables)	13
Numeric columns	5
Categorical (text) columns	8
Target variable	Loan_Status (Y / N)

Each row represents one loan application. Columns cover applicant identity (Loan_ID), demographics (Gender, Married, Dependents, Education, Self_Employed), financials (ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term), credit standing (Credit_History), and location (Property_Area).

4. Data Types

Column	Dtype	Column	Dtype
Loan_ID	object	CoapplicantIncome	float64
Gender	object	LoanAmount	float64
Married	object	Loan_Amount_Term	float64
Dependents	object	Credit_History	float64
Education	object	Property_Area	object
Self_Employed	object	Loan_Status	object
ApplicantIncome	int64		

Two data-type issues are worth flagging for the encoding/cleaning stage: Dependents is stored as text (object) rather than numeric because of the “3+” category, and Credit_History, though conceptually a Yes/No flag, is stored as float64 (1.0 / 0.0) because of missing values, both will need explicit handling before modelling.

5. Missing Values

Column	Missing Count	Missing %
Credit_History	50	8.14%
Self_Employed	32	5.21%
LoanAmount	22	3.58%
Dependents	15	2.44%
Loan_Amount_Term	14	2.28%
Gender	13	2.12%
Married	3	0.49%

149 missing cells in total, spread across 7 of the 13 columns (both numeric and categorical). No column exceeds 10% missing, so all can reasonably be imputed rather than dropped. Credit_History has the highest missing rate and is also, as later analysis will confirm, one of the variables most associated with the target, so its imputation strategy deserves particular care.

6. Duplicate Records

A full-row duplicate check (df.duplicated().sum()) returned 0. A separate check on Loan_ID confirmed all 614 identifiers are unique. No duplicate applications are present in the dataset.

7. Summary Statistics – Numeric Variables

Variable	Mean	Std Dev	Min	Median	Max	Skew
ApplicantIncome	5,403	6,109	150	3,813	81,000	6.54
CoapplicantIncome	1,621	2,926	0	1,189	41,667	7.49
LoanAmount (\$k)	146.4	85.6	9	128	700	2.68
Loan_Amount_Term (mo.)	342.0	65.1	12	360	480	-2.36
Credit_History (0/1)	0.84	0.36	0	1	1	-1.88

ApplicantIncome and CoapplicantIncome are both extremely right-skewed (skew > 6), driven by a small number of very high earners, the maximum applicant income (81,000) is over 20x the median (3,813). LoanAmount is also right-skewed. Loan_Amount_Term and Credit_History are left-skewed because they are dominated by a single common value (360 months and 1.0, respectively). These distributions signal a clear need for scaling and outlier treatment before modelling.

8. Target Variable: Loan_Status

Loan_Status	Applications	% of Dataset
Y (Approved)	422	68.7%
N (Rejected)	192	31.3%

9. Categorical Variables – Overview

Variable	Categories	Most Common	Missing
Gender	Male, Female	Male (489)	13
Married	Yes, No	Yes (398)	3
Dependents	0, 1, 2, 3+	0 (345)	15
Education	Graduate, Not Graduate	Graduate (480)	0
Self_Employed	Yes, No	No (500)	32
Property_Area	Urban, Semiurban, Rural	Semiurban (233)	0

10. Summary & Readiness for Preprocessing

The dataset is small (614 rows) but realistic: it has no duplicate applications, but missing values are present across seven columns (up to 8.1% in Credit_History) and several numeric variables are heavily skewed with likely outliers (ApplicantIncome, CoapplicantIncome, LoanAmount). The target variable is moderately imbalanced (68.7% approved vs 31.3% rejected). These findings directly define the preprocessing tasks for the remainder of Week 2: missing-value imputation, outlier treatment, categorical encoding (including the mixed-type Dependents column), and feature scaling ahead of Week 3 model development.

B. Feature Engineering & Data Preprocessing

1. Data Cleaning

1.1 Missing Values

Column	Missing	Strategy	Reason
Gender	13	Mode ('Male')	Categorical; mode preserves the most common category
Married	3	Mode ('Yes')	Categorical; mode preserves the most common category
Self_Employed	32	Mode ('No')	Categorical; mode preserves the most common category
Dependents	15	Mode (0)	Discrete count dominated by a single value (0)
Credit_History	50	Mode (1.0)	Binary flag; 84% of known values are 1.0
Loan_Term_Months	14	Mode (360)	Standard term covers the large majority of loans
Loan_Amount	22	Median (128)	Right-skewed (skew=2.68); median avoids being pulled up by high-value loans

1.2 Duplicate Records

A full-row duplicate check confirmed 0 duplicates (consistent with the Part 2 inspection). The de-duplication step was still included in the pipeline code for repeatability if the dataset is refreshed.

1.3 Data Type Corrections

- Dependents' "3+" category was converted to the integer 3 so the column could be treated as numeric rather than text, since it represents an ordered count.
- Credit_History was kept as a numeric 0/1 flag (its natural representation) after imputation, rather than being treated as free-text categorical.

2. Feature Engineering

2.1 Removed Features

- Loan_ID – a unique identifier with no predictive relationship to loan approval; retaining it risks the model learning spurious row-level noise.

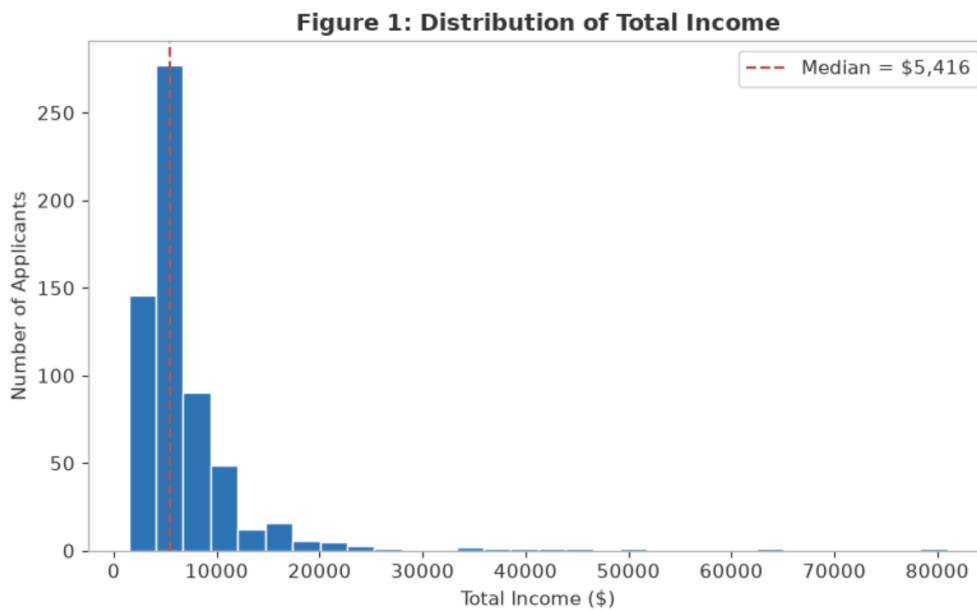
2.2 Renamed Columns

Original	Renamed To
ApplicantIncome	Applicant_Income
CoapplicantIncome	Coapplicant_Income
LoanAmount	Loan_Amount

Renamed for consistent snake_case naming, and to make the unit of Loan_Term_Months explicit rather than implicit.

2.3 New Features Created

Feature	Formula	Business Rationale
Total_Income	Applicant_Income + Coapplicant_Income	Household repayment ability depends on combined income, not the applicant alone
Monthly_Repayment	$(\text{Loan_Amount} \times 1000) / \text{Loan_Amount_Term}$	Approximate monthly instalment, a more direct repayment-burden measure than amount or term alone
Balance_Income	Total_Income - Monthly_Repayment	Proxies the income an applicant has left after covering the estimated loan installment, which is intuitively closer to real repayment capacity than raw income.
Loan_Income_Ratio	$(\text{Loan_Amount} \times 1000) / \text{Total_Income}$	Debt-to-income style ratio; same loan amount means different risk at different income levels



Shows the shape of the Total Income distribution: heavily right-skewed, with most applicants earning under \$10,000 and a long tail of higher earners. The dashed line marks the median (\$5,416), which sits well below the mean, confirming the skew.

3. Outlier Detection & Handling

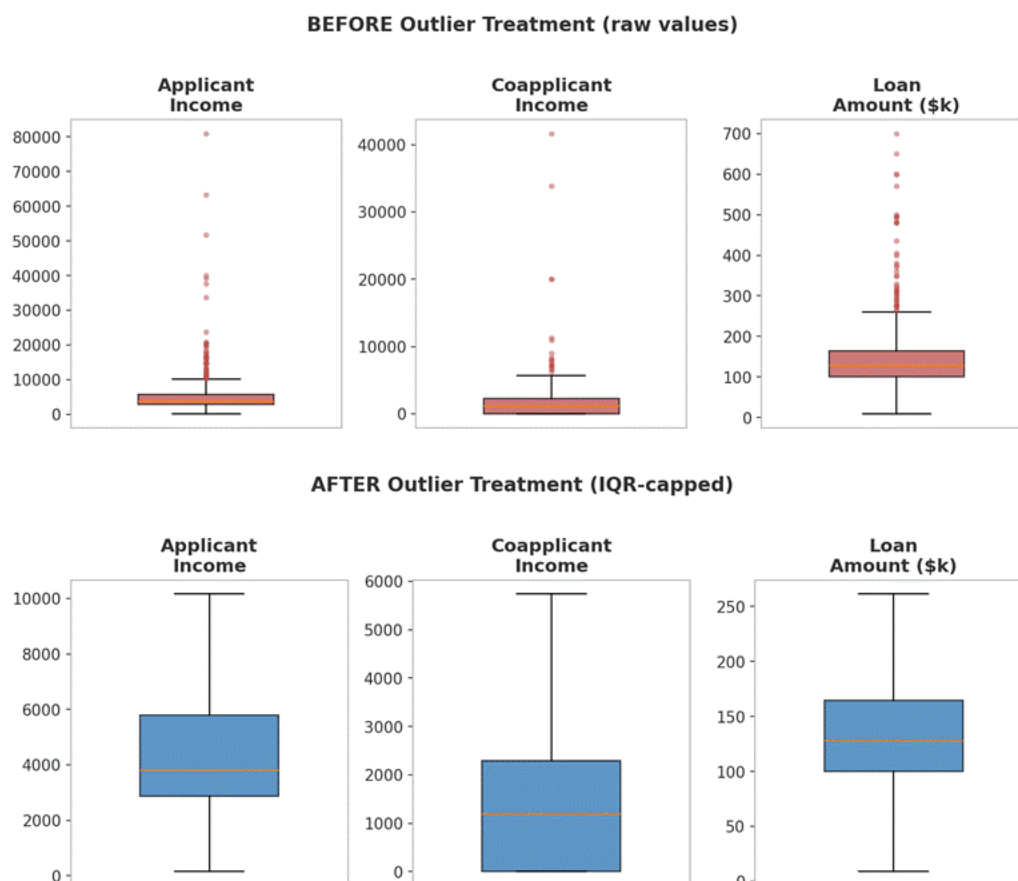


Fig. 2.: Boxplots were used to visually inspect the continuous financial variables before deciding on a numerical detection method.

3.1 Method: IQR, not Z-score

The IQR (Interquartile Range) method was chosen over Z-score. Z-score assumes an approximately normal distribution; these variables are heavily right-skewed (Applicant_Income skew = 6.54, Coapplicant_Income skew = 7.49), so Z-score would under-flag genuine high-end outliers and can misbehave near a compressed mean. The IQR method makes no normality assumption, which better suits this data.

3.2 Treatment: Capping, not Removal

Outliers were capped (winsorised) at the IQR fences ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$) rather than removed as rows. With only 614 records, deleting every flagged row would discard legitimate high-income or high-loan-amount applicants and shrink an already limited training set. Capping keeps every row while limiting the influence of extreme values.

Column	Outliers Capped	Capped Range
Applicant_Income	50	[0, 10,171]

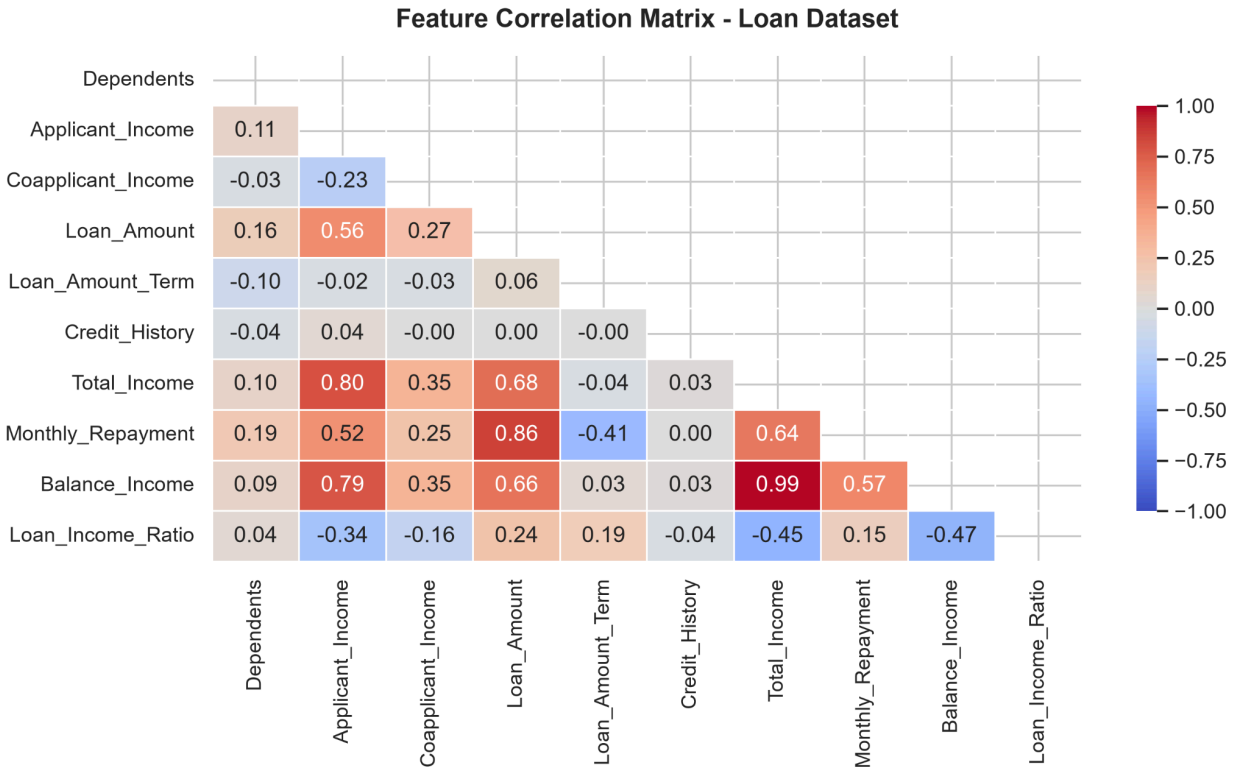
Coapplicant_Income	18	[0, 5,743]
Loan_Amount	41	[3.5, 261.5]
Total_Income	50	[0, 12,555]
Monthly_Repayment	50	[0, 831]
Balance_Income	52	[0, 11,688.6]
Loan_Income_Ratio	25	[5.8, 41.6]

Negative lower bounds produced by the IQR formula were floored at 0, since income and loan values cannot be negative

4. Correlation Check (Redundant Features)

Feature Pair	Correlation	Multicollinearity Risk	Action Taken / Rationale
Total_Income ↔ Balance_Income	0.99	High	Drop Balance_Income
Loan_Amount ↔ Monthly_Repayment	0.86	High	Drop raw Loan_Amount Retain Monthly_Repayment as it better captures monthly cash flow burden.
Applicant_Income ↔ Total_Income	0.80	High	Drop raw Applicant_Income and Coapplicant_Income.
Applicant_Income ↔ Balance_Income	0.79	High	Resolved automatically by dropping Applicant_Income.

Fig.3. : Balance_Income, Loan_Amount, Applicant and Coapplicant Income were dropped due to strong correlation



5. Feature Encoding

5.1 Label Encoding – Binary Variables

Column	Mapping
Gender	Female = 0, Male = 1
Married	No = 0, Yes = 1
Education	Graduate = 0, Not Graduate = 1
Self_Employed	No = 0, Yes = 1
Loan_Status (target)	N = 0, Y = 1

Chosen because each variable has exactly two categories. For a binary variable, label encoding is mathematically equivalent to one-hot encoding with one column dropped, so label encoding is simpler with no loss of information.

5.2 One-Hot Encoding – Property_Area

Property_Area (Urban / Semiurban / Rural) was one-hot encoded into Property_Area_Semiurban and Property_Area_Urban (Rural as the implicit baseline, drop_first=True). This variable is nominal with no natural order; label encoding it would wrongly imply Rural < Semiurban < Urban (or similar), so one-hot encoding was used to keep each category independent. Dropping the first category avoids the dummy-variable trap (perfect multicollinearity between the encoded columns).

5.3 Left As-Is

Dependents (0–3, an ordinal count) and Credit_History (already a clean 0/1 flag) required no further encoding.

6. Feature Scaling

RobustScaler was chosen over StandardScaler and MinMaxScaler. RobustScaler centres on the median and scales by the IQR, rather than the mean/standard deviation (StandardScaler) or the min/max range (MinMaxScaler). Even after capping, the continuous financial variables remain right-skewed, so StandardScaler would still be pulled by the residual skew via the mean, and MinMaxScaler is highly sensitive to the capped boundary values. RobustScaler gives the most stable, outlier-resistant result for this dataset.

Scaled	Not Scaled (and why)
Loan_Amount_Term, Total_Income, Monthly_Repayment, Loan_Income_Ratio	Dependents (small-range ordinal, 0–3) and Credit_History (already clean 0/1) , scaling adds no value and reduces interpretability

7. Feature Selection

7.1 Correlation with the Target Variable

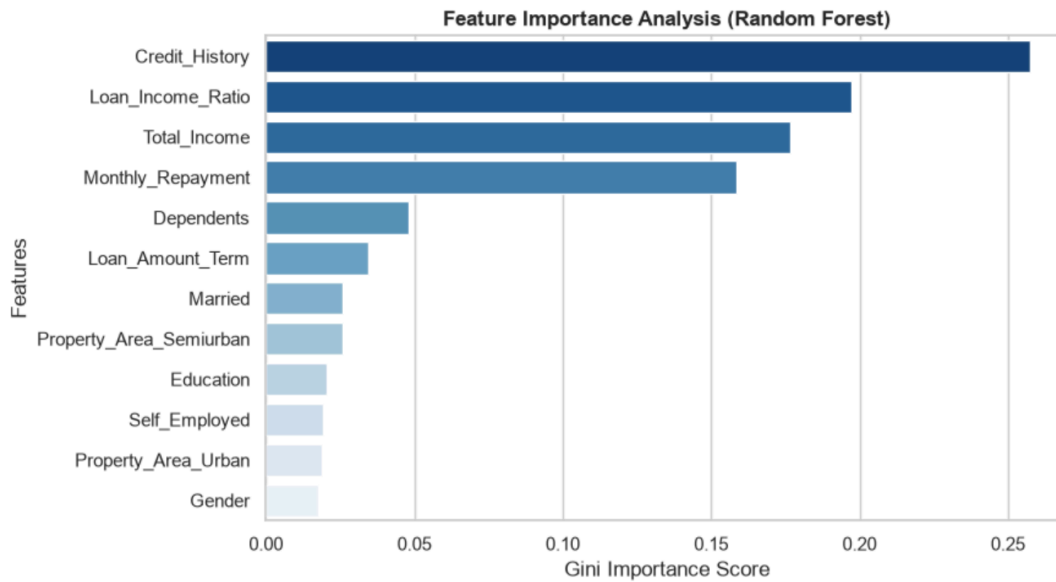
Feature	Correlation with Loan_Status
Credit_History	0.541
Property_Area_Semiurban	0.14
Married	0.091
Education	−0.086
Loan_Income_Ratio	−0.063
Monthly_Repayment	−0.049
Other features	all below ± 0.05

Credit_History stands far above every other feature in raw correlation with the target, more than 3.5x the next-strongest correlation. This single variable is the clearest linear signal in the dataset.

7.2. Feature Importance Analysis

A Random Forest classifier (300 trees, max depth 6) was trained on all 13 features to rank them by Gini importance score, cross-checked against Mutual Information scores (a model-free measure of dependency that also captures non-linear relationships).

Fig. 4: Credit_History dominates feature importance



Rank	Feature	RF Importance	Mutual Info
1	Credit_History	0.426	0.138 (highest)
2	Loan_Income_Ratio	0.106	0.000
3	Total_Income	0.097	0.000
4	Monthly_Repayment	0.060	0.000

Both methods agree Credit_History dominates. Random Forest ranks the engineered income/ratio features next, reflecting their role in capturing non-linear interactions the model exploits; Mutual Information (which measures simpler dependency) rates most other features near zero, suggesting their individual signal is weak in isolation even though they may still contribute through interactions the tree-based model can exploit.

7.3 Features Confirmed as Most Important

- **Most important:** Credit_History: the clear standout by every method used (correlation, RF importance, mutual information).
- Loan_Income_Ratio and Total_Income: The strongest of the engineered financial features, confirming the Part 3 feature engineering added real predictive value beyond the raw income columns.
- Monthly_Repayment: Secondary but consistent signal around loan size and repayment burden.

8. Final Machine-Learning-Ready Dataset

Property	Value
Rows	614
Columns	13 (12 features + target)

Missing values	0
Duplicate rows	0
Categorical columns remaining	0 (all encoded)
File	loan_prediction_ml_ready.csv

More Visualizations

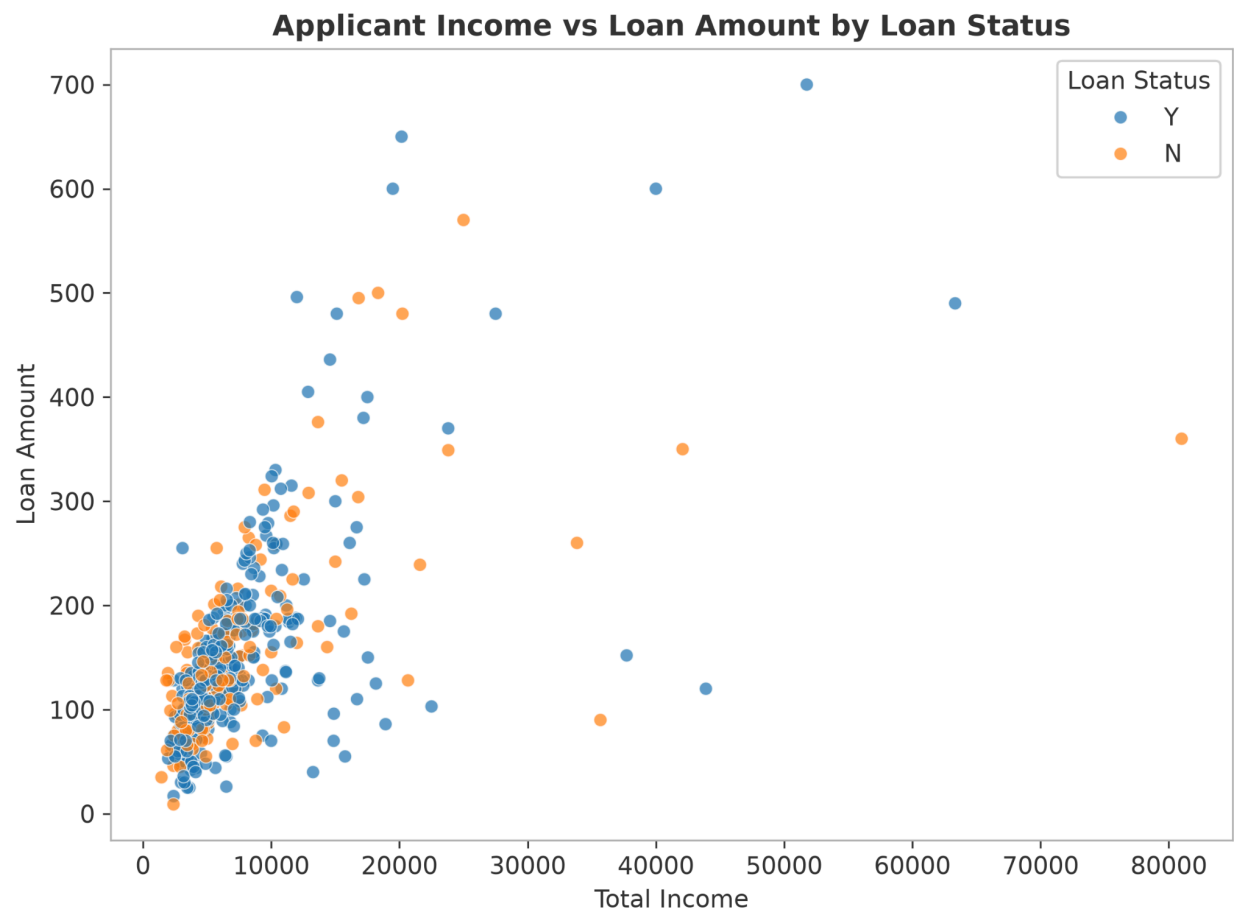


Fig. 5: Examines the relationship between total income and loan amount while distinguishing approved and rejected applications.

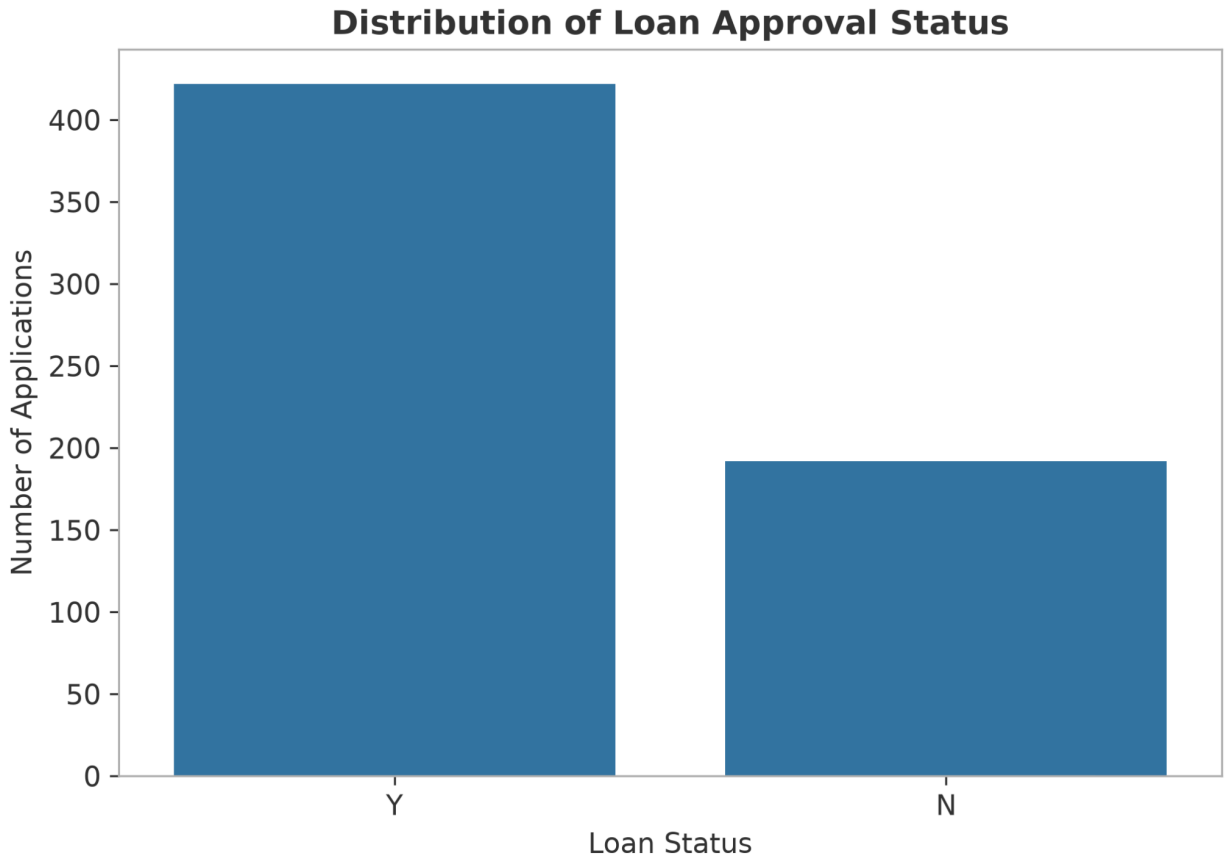


Fig.6.: The dataset is imbalanced due to more approved than rejected loan applications.

Answers to Business Questions

Q1. Which features are most relevant to the prediction problem?

Short Answer:

Credit_History is by far the most relevant feature (correlation with Loan_Status = 0.541, Random Forest importance = 0.426, Mutual Information = 0.138, all far above any other variable). The engineered financial features Loan_Income_Ratio and Total_Income rank next in Random Forest importance. Property_Area_Semiurban and Married show the next-strongest raw correlation with the target after Credit_History.

Q2. Which variables require encoding?

Short Answer:

Five binary categorical variables (**Gender, Married, Education, Self_Employed, and the target Loan_Status**) were label encoded, since label encoding a two-category variable is mathematically equivalent to one-hot encoding with one column dropped. Property_Area (3 nominal categories: Urban, Semiurban, Rural) was one-hot encoded rather than label encoded, because it has no natural order and label encoding would wrongly imply one (e.g. Rural < Semiurban < Urban).

Q3. Which variables require scaling or normalization?**Short Answer:**

The continuous financial variables required scaling: Loan_Amount_Term, Total_Income, Monthly_Repayment, and Loan_Income_Ratio. RobustScaler (median/IQR-based) was used rather than StandardScaler or MinMaxScaler, because these variables remain right-skewed even after outlier capping, and RobustScaler is the most resistant of the three to residual skew and extreme values. Dependents (a small ordinal range, 0–3) and Credit_History (already a clean 0/1 flag) were left unscaled, since scaling adds no value to variables that are already small-range or binary.

Q4. Are there any redundant or highly correlated features?**Short Answer:**

Yes. Two redundancies were found and resolved. First, an engineered feature, Balance_Income (Total_Income – Monthly_Repayment), was found to be 0.989-correlated with Total_Income – a near-perfect linear combination of two features already present – and was dropped. Second, Applicant_Income and Coapplicant_Income were found to be subsumed by Total_Income (their sum, correlation 0.796 and 0.45 respectively) and ranked below it in importance, so both were removed at the feature selection stage.

Q5. How should missing values and outliers be handled?**Short Answer:**

Missing values (149 cells across 7 columns, up to 8.1% in Credit_History) were imputed rather than dropped: **mode imputation** for categorical/discrete variables (Gender, Married, Self_Employed, Dependents, Credit_History, Loan_Term_Months) and **median imputation** for the skewed Loan_Amount, since the median is not pulled upward by high-value loans the way the mean would be.

Outliers in the continuous financial variables were detected using the IQR method (chosen over Z-score, since these variables are heavily right-skewed and Z-score assumes normality) and were capped at the IQR fences rather than removed as rows, to avoid discarding legitimate high-income or high-loan-amount applicants from an already small (614-row) dataset.

Q6. What preprocessing technique improved the dataset quality?

Answer:

- 1. Missing value imputation:** Mode/median imputation – eliminated all 149 missing values without discarding any of the 614 rows.
- 2. Data type correction:** Converted “3+” in Dependents to a numeric 3, and clarified Credit_History/Loan_Term_Months as clean numeric fields.
- 3. Feature engineering:** Total_Income, Monthly_Repayment, and Loan_Income_Ratio captured repayment-capacity signals that the raw income/loan columns alone did not fully express, and ranked among the most important features.
- 4. Outlier capping:** IQR capping brought extreme values (e.g. an \$81,000 income outlier) into a realistic range without losing any applicant records.
- 5. Encoding:** Converted all categorical variables into a fully numeric format usable by machine learning algorithms.
- 6. Scaling:** RobustScaler placed all continuous features on a comparable scale, which matters for distance-based and gradient-based ML algorithms in Week 3
- 7. Redundancy removal:** Removing Balance_Income, Applicant_Income, and Coapplicant_Income reduced multicollinearity risk without losing predictive signal.

Q7. Is the dataset ready for machine learning?

Short Answer:

Yes. The final dataset (loan_prediction_ml_ready.csv) has 614 rows and 13 columns (12 features + target), zero missing values, zero duplicate rows, zero remaining categorical (non-numeric) columns, no known extreme outliers, and no feature pairs above 0.86 correlation among the retained variables. Every preprocessing decision – cleaning, feature engineering, encoding, scaling, and feature selection – has been documented with its rationale across the Part 2–4 reports, so the dataset is both technically ready and auditable ahead of Week 3 model development.